

# Harshavardhan Hajeri

[GitHub](#) [Portfolio](#) [LinkedIn](#) [tau.harsha@gmail.com](mailto:tau.harsha@gmail.com)

## Education

**Birla Institute of Technology and Science, Pilani**

Oct 2021 – June 2026

*M.Sc. (Hons.) Physics | B.E. Mechanical Engineering (Dual Degree)*

## Experience

**Research Staff** | Indian Institute of Technology, Madras

June 2025 – Present

- Conducting research under Prof. Anil Prabhakar; built a **Spatial Photonic Ising Machine** using a phase-only SLM and Fourier optics to solve NP-hard optimization problems via analog interference, with a Metropolis simulated annealing feedback loop and Gaussian beam compensation.
- Developed an SLM-based **holographic modal decomposition** framework for a bow-tie SHG cavity; introduced a digital knife-edge alignment technique and crosstalk matrix calibration, improving average modal self-overlap from 0.86 to 0.96 (fundamental mode: 0.996).
- Implemented **Pound-Drever-Hall (PDH) locking** for a bow-tie SHG cavity and a mode-cleaner cavity, using a Red Pitaya-based lock-in/PID controller driving a PZT through a custom inverting HV amplifier for stable lock acquisition.

**Research Intern** | CeNSE, Indian Institute of Science, Bengaluru

June – August 2024

- Designed a cryostat dipstick in Fusion 360 for low-temperature transport measurements and built a PyQt5/QCoDeS GUI for automated I-V characterization under Dr. Dhavala Suri.

**Quantum Computing Lead** | Google Developer Student Club, BITS Goa

Aug 2023 – May 2025

- Led workshops and study groups reaching 100+ students; organized IBM-sponsored Qiskit Fall Fest 2023 and received the 2024 Qiskit Fall Fest Mentor Badge.

## Projects

**Discrete-Time Quantum Walk** | *Qiskit*

- Implemented coined and split-step DTQW circuits on Qiskit for state preparation and RL-based optimization; demonstrated advantage over classical random walk baselines on the N-Armed Bandit problem.

**Entanglement in Cluster States** | *Strawberry Fields*

under Dr. Sanjib Dey, BITS Goa

- Simulated 2D cluster states and computed entanglement measures on photon-subtracted squeezed vacuum states in continuous-variable quantum optics.

**MEMS Microphone Simulation** | *COMSOL Multiphysics*

under Dr. D. G. Patil, BITS Goa

- Performed structural-acoustic coupled simulation of a capacitive MEMS microphone; proposed design modifications that improved simulated sensitivity.

## Skills

**Programming & Frameworks:** Python, Qiskit, QuTiP, PennyLane, Strawberry Fields, JavaScript (React), L<sup>A</sup>T<sub>E</sub>X

**Tools & Platforms:** Git/GitHub, MATLAB, LabVIEW, Simulink, COMSOL, FINESSE 3, Red Pitaya, IBM Quantum, Fusion 360, Jupyter Notebook

## Teaching Experience

**Teaching Assistant** | *Quantum Information & Computation, BITS-F386*

Jan – May 2024

- Under Dr. Radhika Vathsan - designed and graded assignments for 50 students; created Qiskit-based course content and Jupyter Notebook tutorials.

**Mentor** | *Quantum Computing with Qiskit (Quark STP) | Theory of Relativity (CTE)*

2022 – 2023

- Mentored 150+ students across two programmes; developed Manim-based materials covering Grover's Algorithm and the foundations of special relativity.

## Conference Presentations

G. Patil\*, **H. Hajeri\***, S. P. Amrithraj, A. Prabhakar. *Modal Decomposition of Cavity SHG Fields using Spatial Light Modulator*. EOP 2025.

**H. Hajeri\***, N. Vinod P.M., G. Patil, S. P. Amrithraj, A. Prabhakar. *Spatial Photonic Ising Machine using Spatial Light Modulators*. EOP 2025.

## Awards & Extra-Curriculars

Seed Fund Grant INR 5,00,000 (PIEDS) | Prof. Suresh Ramaswamy Award INR 40,000 | IBM Qiskit Fall Fest Mentor Badge, 2024

Steering Member, QIndia | Core Member, Nirmaan NGO, BITS Goa | Core Member, Mime Club BITS Goa